B E N C H M A R K R E P O R T

They Saw the Problem And Chose It Anyway

When 12 Models Discussed UV Degradation Then Selected the UV-Vulnerable Material

Behavioral Analysis of 22 LLMs on an Outdoor Piping Material Selection Question

22	27.3%	12	0/4
LLMs Tested	Reported Pass Rate	Chose FRP (Trap)	Qwen Family

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1. Executive Summary

I posed a material selection question to 22 LLMs: design outdoor piping for an odor collection system in Ceyhan, Adana — a location with extreme heat (45–50°C), intense UV radiation, and corrosive H_2S gas. The correct answer is stainless steel 316, which resists all three environmental stressors. The trap is FRP (Fiber Reinforced Plastic), which excels at H_2S resistance but degrades under prolonged UV exposure.

The reported pass rate is 27.3% (6/22). This is the third-lowest in the series, placing it firmly in the hard tier alongside Biogas (14%) and Walkway (36%). After eleven benchmarks, no model has passed every question. The cumulative funnel: 22 → 17 → 15 → 2 → 0 → 0 → 0 → 0 → 0 → 0 → 0.

F1 — The FRP Trap Caught 12 Models: Twelve models chose C (FRP) — the largest single-option trap capture in the entire series. FRP's excellent H_2S resistance made it the most attractive wrong answer. This is the second time FRP has served as a trap in the series (after Walkway #5), but for an entirely different reason: there, the issue was brittle failure at height; here, it is UV degradation under extreme sunlight.

F2 — They Saw the Problem and Chose It Anyway: Nearly all 12 FRP-choosing models discussed UV degradation in their reasoning. They acknowledged that FRP is sensitive to UV, then dismissed the concern by citing UV-resistant resins, gel coats, or protective coatings. They did not miss the hazard — they identified it, evaluated it, and judged it manageable. This is a new failure mode: not ignorance, not partial knowledge, but incorrect judgment after correct identification.

F3 — Qwen: 0/4 — First Complete Family Failure: All four Qwen models chose FRP. This is the first time Qwen has scored 0% on any benchmark. The family that maintained 100% through benchmarks #7–#8 has now failed completely on a material selection question.

F4 — Gemini Splits Again — Old Passes, New Fails: Gemini 2.0 Flash and 2.0 Flash Lite chose E (correct). Gemini 2.5 Flash, 2.5 Pro, 3 Flash Preview, and 3 Pro Preview all chose C (FRP). The older generation passed; the newer generation failed. This is the third benchmark where generational patterns have appeared in Gemini, and the second time the older models outperformed the newer ones.

F5 — R1's First Actual Wrong Answer: For the first time in the series, DeepSeek R1 chose the wrong answer: C (FRP). In all previous benchmarks, R1's failures were parsing artifacts — the model knew the correct answer but formatting prevented it from being scored. Here, R1 genuinely selected the wrong material. Additionally, R1 used a new formatting pattern (**Answer: C**), causing the parser to read "A" from "Answer" — a wrong parse of a wrong answer.

2. The Question

The scenario presents an outdoor piping design challenge in an extreme climate. The key constraint is the combination of three environmental stressors: corrosive H_2S gas, extreme UV radiation (direct sunlight at 45–50°C), and large thermal cycles (5°C to 50°C). No single polymer satisfies all three for a 10-year service life.

The trap exploits the dominance of H_2S resistance in the decision framework. When models evaluate options for a wastewater odor collection system, H_2S resistance naturally becomes the primary criterion. FRP excels at this — and its H_2S performance is so well-documented in training data that it anchors models on the chemical resistance dimension while the UV and thermal dimensions receive secondary consideration.

With 5 options, the random correct answer probability is 20%.

Table 1: The Five Options

Option	Material	Rating	Key Issue
A	PVC	Poor	Degrades under UV and extreme heat. Becomes brittle in prolonged sunlight. Not suitable for outdoor high-temperature applications.
B	Galvanized Steel	Poor	H_2S corrodes the zinc coating rapidly. Short service life in corrosive gas environments.
C	FRP	TRAP	Excellent H_2S resistance but UV degrades the resin matrix. Real-world failures documented within 3–5 years in outdoor installations without maintenance.
D	HDPE	Partial	Good H_2S resistance but softens at high temperatures and has high thermal expansion. UV resistance requires additives.
E	SS 316	CORRECT	Resists H_2S, UV, and thermal cycling. Higher initial cost but meets 10-year service life without degradation.

Table 1: FRP dominates on the H_2S dimension but fails on the UV dimension. The question tests whether models can weigh multiple environmental stressors simultaneously rather than anchoring on the most chemically salient one.

3. Results

Figure 1: Answer Distribution

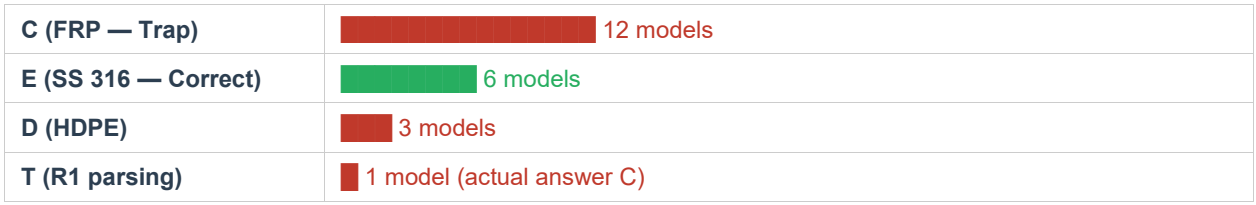

Figure 1: FRP attracted more than half of all models. No model chose A (PVC) or B (Galvanized Steel). The H_2S resistance dimension dominated decision-making.

Table 2: Complete Results — All 22 Models

Model	Ans	Status	Out Tok	Latency	Family

Claude Haiku 4.5	E	PASS	340	3.8s	Claude
Claude Opus 4.1	C	FAIL	330	11.9s	Claude
Claude Opus 4.5	C	FAIL	412	12.7s	Claude
Claude Sonnet 4	C	FAIL	318	8.1s	Claude
Claude Sonnet 4.5	E	PASS	398	10.8s	Claude
Gemini 2.0 Flash	E	PASS	759	5.9s	Gemini
Gemini 2.0 Flash Lite	E	PASS	280	2.9s	Gemini
Gemini 2.5 Flash	C	FAIL	468	12.3s	Gemini
Gemini 2.5 Pro	C	FAIL	505	25.3s	Gemini
Gemini 3 Flash Preview	C	FAIL	315	8.6s	Gemini
Gemini 3 Pro Preview	C	FAIL	405	21.8s	Gemini
Gemma 3 12B	C	FAIL	~370	10.2s	Gemma
Gemma 3 4B	E	PASS	~350	9.2s	Gemma
Gemma 3 27B	D	FAIL	~350	11.9s	Gemma
Gemma 3 1B	D	FAIL	308	6.1s	Gemma
DeepSeek R1 (CoT)	C*	FAIL	1,985	13.1s	DeepSeek
DeepSeek V3.1	D	FAIL	154	2.7s	DeepSeek
DeepSeek V3.2	E	PASS	139	11.0s	DeepSeek
Qwen3 235B A22B Instruct	C	FAIL	281	3.0s	Qwen
Qwen3 Coder 480B	C	FAIL	263	2.6s	Qwen
Qwen3 Next 80B Instruct	C	FAIL	238	3.6s	Qwen
Qwen3 Next 80B Thinking	C	FAIL	2,584	14.4s	Qwen

*Table 2: *R1 answered C (FRP — wrong) inside a think block formatted as **Answer: C**. Parser read "A" from "Answer." This is R1's first genuine wrong answer — previous failures were parsing artifacts of correct answers.*

4. LLM Behavior Analysis

4.1 The FRP Trap Returns — Same Material, Different Failure Mode

FRP has now served as a trap in two benchmarks, making it the only material to appear twice in the trap taxonomy. In the Walkway benchmark (#5), FRP was wrong because of its brittle failure mode at height — a structural concern. Here, FRP is wrong because of UV degradation of the resin matrix — an environmental durability concern. The material's strengths are real: excellent chemical resistance, lightweight, easy to install. But in both cases, models anchored on these salient properties while underweighting a critical weakness specific to the installation context.

Twelve of 22 models chose FRP — the largest single-option capture in the series. By comparison, the Aerobic Trap in Biogas (#4) caught 15 models across multiple wrong options; here, 12 models converged on a single wrong answer. The FRP trap's effectiveness appears to

stem from the material's genuine excellence on the most salient criterion (H_2S resistance), which dominates the decision framework and marginalizes secondary criteria.

4.2 Identified, Evaluated, Dismissed — A New Failure Mode

The most striking behavioral finding on this benchmark is not that models chose FRP, but how they chose it. Nearly all 12 FRP-selecting models explicitly discussed UV degradation in their reasoning. They did not miss the problem. They identified it, acknowledged it, and then dismissed it.

The dismissal followed a consistent pattern: "FRP is sensitive to UV, but UV-resistant vinyl ester resins are available." "Gel coats can protect against UV degradation." "With proper resin selection, UV is manageable." The models treated UV degradation as a solvable engineering detail rather than a fundamental material limitation for a 10-year outdoor installation in extreme sunlight.

This is behaviorally distinct from the failure modes documented in previous benchmarks. In the Tool Selection benchmark (#8), failing models did not identify the hazard at all. In the SCADA Cabling benchmark (#10), failing models identified the problem but applied the wrong solution (partial knowledge). Here, the models identified the problem, proposed a mitigation, and judged it sufficient — but that judgment was wrong. The knowledge was complete; the evaluation was incorrect.

This progression — from missing the hazard, to knowing the problem but not the solution, to knowing everything but judging incorrectly — suggests that LLM failure modes exist on a spectrum of knowledge completeness. At each level, the failure becomes harder to detect: a model that says "FRP has UV issues, but gel coats solve this" sounds more authoritative than one that says "I recommend FRP for its chemical resistance." The more the model knows, the more convincing its wrong answer becomes.

Table 3: Failure Mode Spectrum Across the Series

Failure Mode	Example	What Happened	Detectability
Hazard Blindness	Tool Selection (#8)	Models did not mention spark risk at all	Easy to detect — absence of relevant reasoning
Partial Knowledge	SCADA Cabling (#10)	Models identified EMI but applied wrong solution	Moderate — correct problem, wrong fix
Judgment Error	Odor Pipe (#11)	Models identified UV risk, proposed mitigation, judged it sufficient	Hard to detect — reasoning sounds complete and authoritative

Table 3: The failure mode spectrum. As models' knowledge increases, their wrong answers become more convincing and harder to identify as errors.

4.3 R1's First Genuine Wrong Answer — And a New Parsing Pattern

For the first time in eleven benchmarks, DeepSeek R1 selected the wrong answer. In all previous failures, R1's reasoning was correct but formatting prevented the parser from reading

it. Here, R1 genuinely chose C (FRP), with a 7,489-character think block that discussed FRP's H_2S resistance and concluded it was the best choice. R1 fell for the same trap as 12 other models.

Additionally, R1 introduced a new formatting pattern: **Answer: C** (bold with "Answer:" prefix). The parser's regex stripped non-letter characters and read "A" from the word "Answer" — a parsing failure of a different kind. The result: R1 is scored as choosing A (which no other model chose), when it actually chose C (which 12 models chose). Both the actual answer and the parsed answer are wrong, but for different reasons.

This benchmark marks a behavioral inflection point for R1. Across 10 previous benchmarks, R1 had built a reputation as a model with excellent reasoning but poor formatting. That narrative is now incomplete: R1 can also be wrong.

Table 4: R1 History — Updated With First Wrong Answer

Benchmark	Raw	Answer	Correct?	Issue
#1 Material Selection	<	B	Yes (PASS)	<think> block
#2 Pump RCA	<	A	Yes (FAIL*)	<think> → parsed wrong
#3 Dewatering RCA	<	A	No (FAIL)	<think> + wrong answer
#4 Biogas	—	—	—	(data pending)
#5 Walkway Material	*	B	No (FAIL)	**B** markdown
#6 Confined Space	*	O	Yes (FAIL*)	**O** → parsed wrong
#7 Sample Preservation	<	D	Yes (FAIL*)	<think> → T parsed
#8 Tool Selection	<	D	Yes (FAIL*)	<think> → T parsed
#9 Emergency Flocc.	<	C	Yes (FAIL*)	<think> → T parsed
#10 SCADA Cabling	<	E	Yes (FAIL*)	<think> → T parsed
#11 Odor Pipe	*	C	No (FAIL)	**Answer: C** → A parsed

Table 4: R1 has encountered formatting issues in 9 of 11 benchmarks. For the first time, the actual answer was also wrong — ending the pattern of "correct reasoning, lost to formatting."

4.4 Family Behavioral Patterns — No Safe Haven

Figure 2: Family Pass Rates

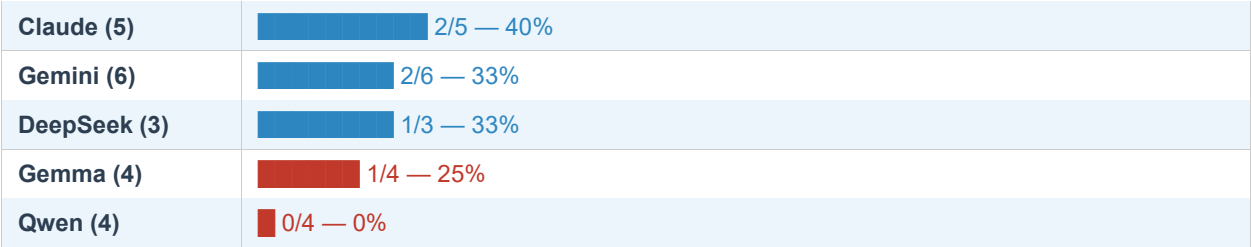

Figure 2: No family exceeds 40%. This is the most evenly distributed failure across families in the series — no safe haven, no dominant winner.

This is the first benchmark where no family scored above 40%. In every previous benchmark, at least one family achieved 67% or higher. The FRP trap's effectiveness was cross-family: it caught Claude, Gemini, Gemma, DeepSeek, and Qwen models alike. H_2S-resistance anchoring appears to be a universal cognitive shortcut rather than a family-specific vulnerability.

Gemini's generational split continues its now-established pattern. The 2.0 generation (Flash and Flash Lite) passed; the 2.5 and 3.x generation failed. This is the same direction as the SCADA Cabling benchmark (#10), but the opposite of Tool Selection (#8) where newer models outperformed older ones. Across three benchmarks with generational splits, older Gemini models have a 2:1 advantage over newer ones — a pattern that defies the "newer is better" expectation.

Qwen's 0/4 is the first complete family failure for Qwen in the series. All four models — including the 235B instruct model that passed 8 of the previous 10 benchmarks — chose FRP. The trap's anchoring on H_2S resistance was strong enough to override Qwen's generally strong domain knowledge.

4.5 Trap Taxonomy Update — Eleven Benchmarks

Table 5: Updated Trap Taxonomy

Benchmark	Trap Name	The Shortcut	Pass %	What Was Missed
#1 Material Selection	—	—	100%	
#2 Pump RCA	Electrical Trap	High current = winding	77%	Mechanical overload
#3 Dewatering RCA	Local Trap	Vibration = bearings	68%	Upstream grit
#4 Biogas	Aerobic Trap	SOB = aerobic → aeration	14%	Digester sulfur cycling
#5 Walkway	Corrosion Trap (FRP — Structural)	H_2S = corrosion resist.	36%	Brittle failure mode
#6 Confined Space	"Almost Safe"	Protocol = sufficient	68%	Engineer's presence
#7 Sample Preserv.	Acid Trap	Acid = always correct	95.5%	Unknown → general preserv.
#8 Tool Selection	"False Safe"	No gas = safe	68.2%	Trapped gas in sludge
#9 Emergency Flocc.	Coagulant Trap	Coagulant = flocculant	68.2%	Mechanism confusion
#10 SCADA Cabling	Cost-Reliability	Separate cables = enough	63.6%	Physical separation
#11 Odor Pipe	UV Anchoring (FRP — UV)	H_2S resist. = best material	27.3%	UV degradation over 10 years

Table 5: Eleven benchmarks. FRP appears as a trap material for the second time — the only material to feature twice, each time for a different failure mechanism.

5. Cross-Benchmark Comparison — Eleven Questions Deep

Figure 3: Pass Rate Across 11 WWTP Benchmarks

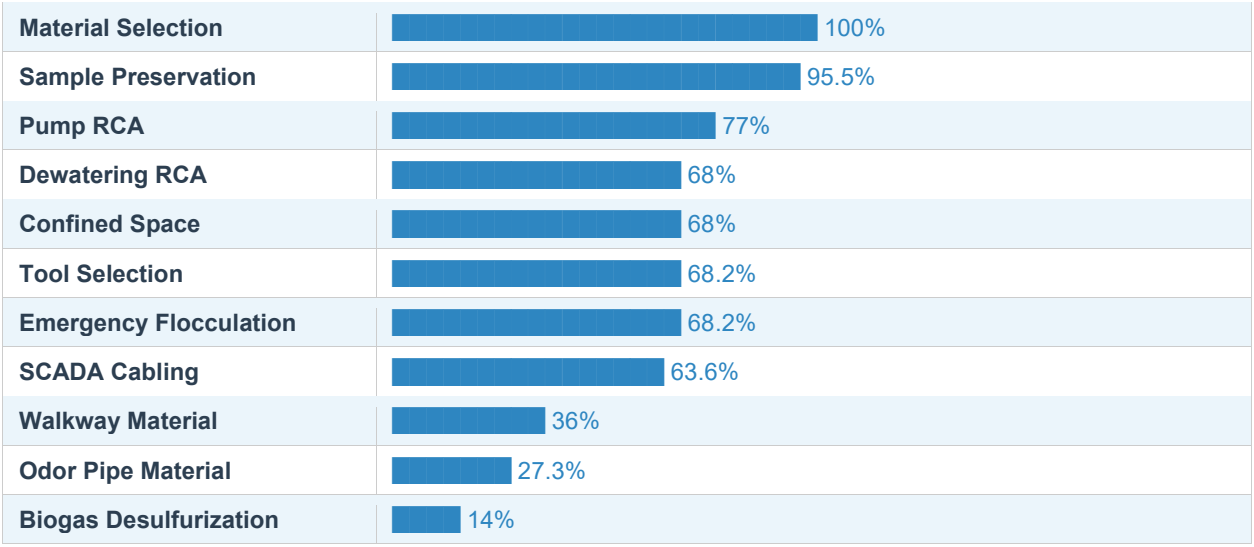

Figure 3: Eleven benchmarks spanning 14% to 100%. Odor Pipe enters the hard tier at 27.3%, the third-lowest in the series.

The difficulty gradient now spans three clear tiers with eleven data points. The hard tier (below 40%) now includes three benchmarks: Biogas at 14%, Odor Pipe at 27.3%, and Walkway at 36%. All three hard-tier benchmarks share a common characteristic: they require weighing multiple competing material or process properties simultaneously, where the most salient property is not the decisive one.

The cumulative funnel remains at zero for the seventh consecutive benchmark: 22 → 17 → 15 → 2 → 0 → 0 → 0 → 0 → 0 → 0 → 0. No model has passed all eleven questions. The funnel closed at benchmark #5 and has remained at zero through seven additional benchmarks of varying difficulty — from the easiest in the series (Sample Preservation at 95.5%) to one of the hardest (this benchmark at 27.3%). This persistence across such different difficulty levels suggests the closure is structural, not coincidental. With this benchmark, every family has now experienced at least one complete or near-complete failure: Claude at 40% (#10, #11), Gemini at 33% (#5, #11), Gemma at 0% (#8) and 25% (#11), DeepSeek at 0% (#6) and 33% (#11), and Qwen at 0% (#11). No family is immune.

6. Conclusions

This benchmark adds an eleventh dimension to the WWTP series and reveals a failure mode that may be the most difficult to detect: judgment error with complete knowledge. Models that identified UV degradation as a risk, proposed mitigations, and chose FRP anyway are harder to catch than models that never considered UV at all. The failure mode spectrum — from hazard blindness through partial knowledge to judgment error — is an emerging framework for categorizing LLM reliability in advisory contexts.

The FRP trap's second appearance in the series confirms that material-specific anchoring is a persistent cognitive pattern. In Walkway (#5), FRP was wrong for structural reasons; here, for durability reasons. The common thread is that FRP's strong chemical resistance acts as a cognitive anchor that overshadows context-specific weaknesses. This suggests that certain materials, chemicals, or solutions that are well-represented in training data may consistently bias model recommendations regardless of the specific installation context.

R1's first genuine wrong answer ends a narrative that had persisted for ten benchmarks. The model that was known for "correct reasoning, lost to formatting" can now also be wrong on substance. This does not diminish R1's generally strong reasoning capability — it contextualizes it. Like every other model in the series, R1 has blind spots.

The 27.3% pass rate places this benchmark in the hard tier, alongside Biogas and Walkway. The three hard-tier benchmarks now suggest a pattern: questions that require simultaneous multi-criteria evaluation — where the most salient criterion is not the decisive one — produce the lowest pass rates. This is consistent with the anchoring-on-salient-features pattern documented throughout the trap taxonomy. Notably, all three hard-tier benchmarks involve material or process selection where the dominant property in training data (chemical resistance, corrosion resistance, aerobic biology) is not the property that determines the correct answer — suggesting that training data frequency may systematically bias models toward certain criteria over others.

After eleven benchmarks, no model and no family maintains a clean record. The series has demonstrated that domain-specific engineering knowledge in current LLMs is deep but narrow — models may excel on the criterion they anchor on while systematically underweighting criteria that are less prominent in their training data but equally important in the real world.

7. Methodology Notes

Non-determinism: Single run per model. Results are indicative, not absolute.

Single question: This report analyzes one question with 5 options. Random correct answer probability is 20%.

Parsing: DeepSeek R1 formatted its answer as **Answer: C** (bold with prefix). Parser read "A" from "Answer." Unlike previous benchmarks, R1's actual answer (C) was also wrong. No adjusted pass rate is reported for this benchmark as the parsing issue did not affect the overall pass/fail outcome.

Token counts: Some Gemma models reported 0 output tokens in API metrics despite visible responses. Counts marked with ~ are estimated.

API conditions: All models accessed via Kaggle Benchmarks platform API with default temperature settings.

Series consistency: All eleven benchmarks use the same platform, the same first-character parser, and the same default temperature settings.